浙江大学实验报告

姓名：金宣妤　　专业：电子科学与技术　　学号：3240100333

课程名称：Signals and Systems　　任课老师：Prof. Xiaoming Chen

实验名称：Lab1: Introduction to Matlab in Signals and Systems　　实验日期：2026.3.17

1 Experimental Objectives

(1)Learn to represent, manipulate, and analyze basic discrete-time (DT) and continuous-time (CT) signals in the MATLAB environment, and become familiar with vector indexing, matrix operations, and basic plotting functions (such as stem, plot, ezplot).

(2)Visualize the traditional written theories of "Signals and Systems" (such as periodicity, aliasing, linearity, causality, etc.) through computer programming exercises to build a more intuitive and profound understanding.

2 Experimental Content, Code Analysis, and Result Discussion

(All run on Matlab R2025b)

Problem 1

Consider the signal

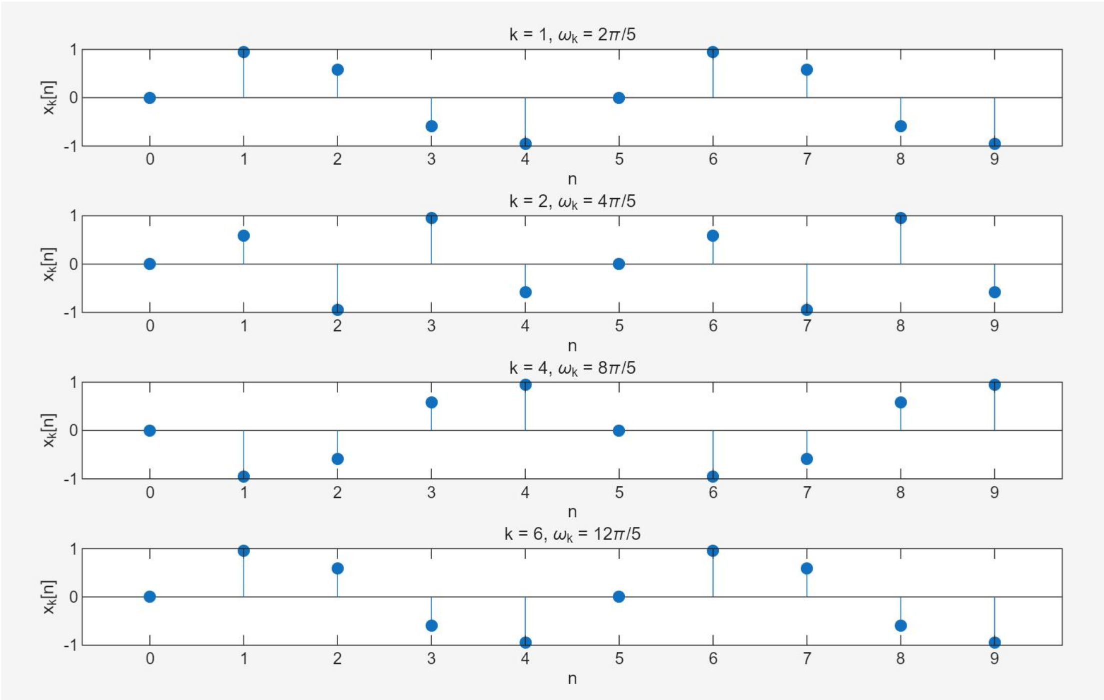
$$x_k[n] = \sin(\omega_k n)\,,$$

where $\omega_k = 2\pi k/5$. For $x_k[n]$ given by $k = 1, 2, 4$, and 6, use **stem** to plot each signal on the interval $0 \le n \le 9$. All of the signals should be plotted with separate axes in the same figure using **subplot**. How many unique signals have you plotted? If two signals are identical, explain how different values of ω_k can yield the same signal.

```
clear; clc;
n = 0:9;
k_vals = [1, 2, 4, 6];

figure;
for i = 1:4
    k = k_vals(i);
    wk = 2*pi*k/5;
    xk = sin(wk * n);

    subplot(4, 1, i);
    stem(n, xk, 'filled');
    title(['k = ', num2str(k), ', \omega_k = ', num2str(2*k), '\pi/5']);
    xlabel('n'); ylabel('x_k[n]');
end
```

Code Logic: Use a for loop to iterate through different values of k, and plot the discrete-time sequences using the stem function.

Experimental Results and Conclusion: When k=1 and k=6, their plots are exactly identical.



Problem 2

Now consider the following three signals

$$x_1[n] = \cos\left(\frac{2\pi n}{N}\right) + 2\cos\left(\frac{3\pi n}{N}\right),$$
$$x_2[n] = 2\cos\left(\frac{2n}{N}\right) + \cos\left(\frac{3n}{N}\right),$$
$$x_3[n] = \cos\left(\frac{2\pi n}{N}\right) + 3\sin\left(\frac{5\pi n}{2N}\right).$$

Assume $N = 6$ for each signal. Determine whether or not each signal is periodic. If a signal is periodic, plot the signal for two periods, starting at $n = 0$. If the signal is not periodic, plot the signal for $0 \leq n \leq 4N$ and explain why it is not periodic. Remember to use `stem` and to appropriately label your axes.

Code Logic:

Signal 1 (x1): Math tells us this signal is periodic (it repeats) with a period of 12. The code creates 24 time steps (n1 = 0:23) to draw exactly 2 full cycles.

Signal 2 (x2): Math tells us this signal is NOT periodic (it never repeats). The code creates 25 time steps (n2 = 0:24) to draw a long line of points, just like the homework asked.

Signal 3 (x3): Math tells us this signal is periodic with a period of 24. The code creates 48 time steps (n3 = 0:47) to draw exactly 2 full cycles.

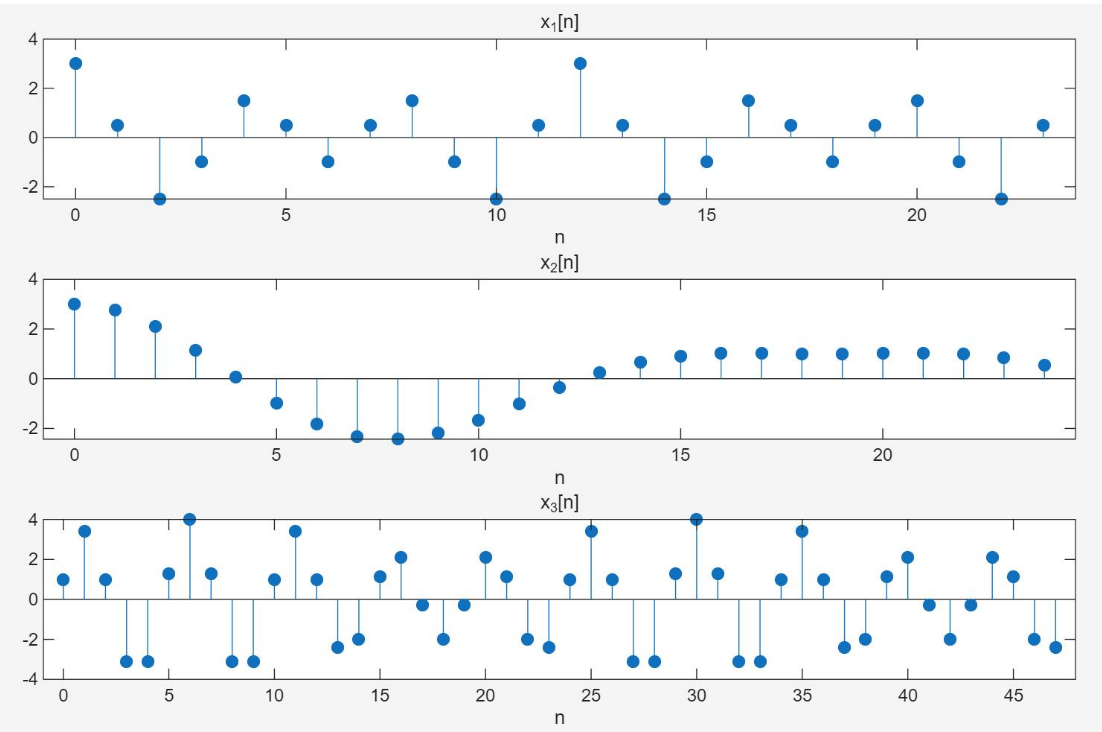
Experimental Results and Conclusion:

x1[n] is periodic. It is made of two cosine waves. The first wave repeats every 6 steps. The second wave repeats every 4 steps. The Least Common Multiple (LCM) of 6 and 4 is 12. This means the whole signal repeats every 12 steps. The graph proves this perfectly.

x2[n] is not periodic. In discrete time, a wave only repeats if its frequency contains the number pi. Because there is no pi in the formula of x2[n], the signal will never perfectly repeat on whole numbers. The graph shows a messy pattern.

x3[n] is periodic. It is made of two waves. The first wave repeats every 6 steps. The second wave repeats every 24 steps. The LCM of 6 and 24 is 24. Therefore, the whole signal repeats every 24 steps. The graph clearly shows one full pattern every 24 dots.

```
clear; clc;
N = 6;

% x1[n]: 周期信号，周期为12。绘制两个周期 (0~23)
n1 = 0:23;
x1 = cos(2*pi*n1/N) + 2*cos(3*pi*n1/N);
figure;
subplot(3, 1, 1); stem(n1, x1, 'filled');
title('x_1[n]'); xlabel('n');

% x2[n]: 非周期信号。绘制 0~4N (0~24)
n2 = 0:24;
x2 = 2*cos(2*n2/N) + cos(3*n2/N);
subplot(3, 1, 2); stem(n2, x2, 'filled');
title('x_2[n]'); xlabel('n');

% x3[n]: 周期信号，周期为24。绘制两个周期 (0~47)
n3 = 0:47;
x3 = cos(2*pi*n3/N) + 3*sin(5*pi*n3/(2*N));
subplot(3, 1, 3); stem(n3, x3, 'filled');
title('x_3[n]'); xlabel('n');
```



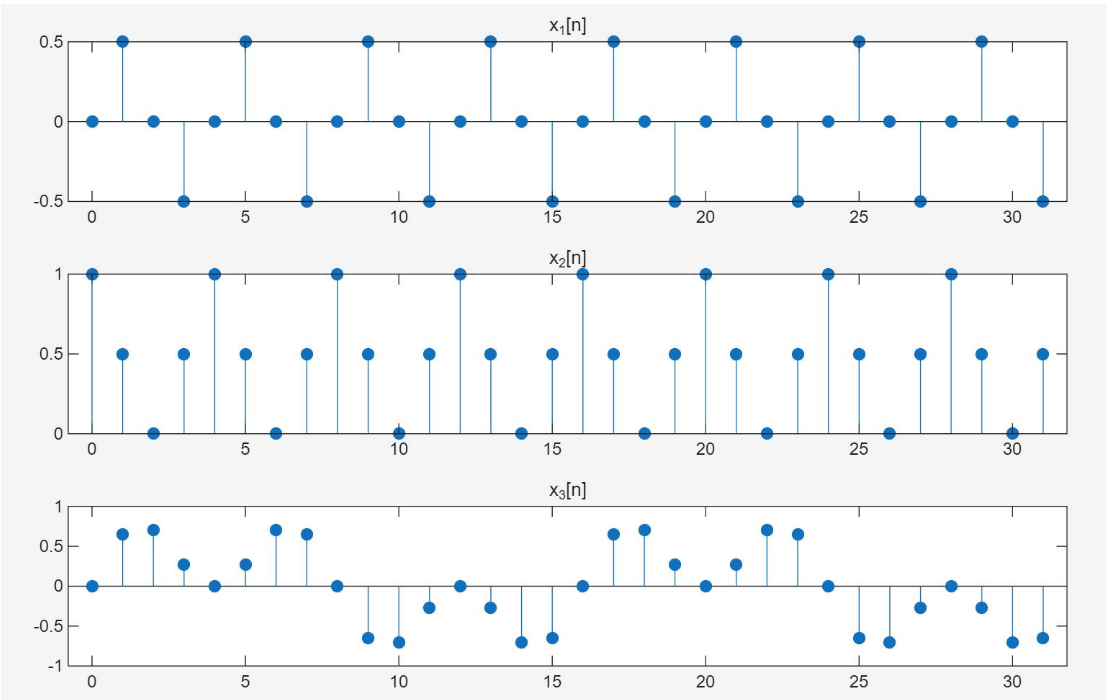
Problem 3

Plot each of the following signals on the interval $0 \le n \le 31$:

$$x_1[n] = \sin\left(\frac{\pi n}{4}\right)\cos\left(\frac{\pi n}{4}\right),$$
$$x_2[n] = \cos^2\left(\frac{\pi n}{4}\right),$$
$$x_3[n] = \sin\left(\frac{\pi n}{4}\right)\cos\left(\frac{\pi n}{8}\right).$$

What is the fundamental period of each signal? For each of these three signals, how could you have determined the fundamental period without relying upon MATLAB?

```
clear; clc;
n = 0:31;
x1 = sin(pi*n/4) .* cos(pi*n/4);
x2 = cos(pi*n/4).^2;
x3 = sin(pi*n/4) .* cos(pi*n/8);

figure;
subplot(3, 1, 1); stem(n, x1, 'filled'); title('x_1[n]');
subplot(3, 1, 2); stem(n, x2, 'filled'); title('x_2[n]');
subplot(3, 1, 3); stem(n, x3, 'filled'); title('x_3[n]');
```



Experimental Results and Conclusion:

The fundamental period of x1[n] is 4. Using the double-angle formula, the signal can be simplified to

$$x_1[n] = \sin(\frac{\pi n}{4})\cos(\frac{\pi n}{4}) = \frac{1}{2}\sin(\frac{\pi n}{2})$$

the fundamental period T1 = 4.

The fundamental period of x2[n] is 4. Using the power-reduction formula, we get

$$x_2[n] = \cos^2(\frac{\pi n}{4}) = \frac{1}{2} + \frac{1}{2}\cos(\frac{\pi n}{2})$$

The fundamental period of x3[n] is 16. Using the product-to-sum formula, the signal can be decomposed into

$$x_3[n] = \sin(\frac{\pi n}{4})\cos(\frac{\pi n}{8}) = \frac{1}{2}\left[\sin(\frac{3\pi n}{8}) + \sin(\frac{\pi n}{8})\right]$$

The fundamental period of the combined signal is the least common multiple of the periods of the two sine components , which is T3= 16.

Problem 4

Consider the signals you plotted in Problem 2 and 3. Is the addition of two periodic signals necessarily periodic? Is the multiplication of two periodic signals necessarily periodic? Clearly explain your answers.

Conclusion: Yes. In the context of discrete-time signals, the addition of two periodic signals is necessarily periodic, and the multiplication of two periodic signals is also necessarily periodic .

Addition: For discrete-time signals, if x1[n] and x2[n] are periodic, their fundamental periods N1 and N2 must be integers. Mathematically, any two positive integers are guaranteed to have a Least Common Multiple (LCM). This means that every LCM(N1, N2) samples, both signals will simultaneously complete an integer number of cycles and return to their original states. Therefore, their sum x1[n] + x2[n] is guaranteed to be periodic. This is verified by Problem 2, which are sums of discrete periodic signals.

Multiplication: By the exact same logic, because the integer periods N1 and N2 share a common multiple LCM(N1, N2), the product sequence will also repeat its values perfectly every LCM(N1, N2) samples. This is demonstrated in Problem 3.

Problem 5

For these problems, you are told which property a given system does not satisfy, and the input sequence or sequences that demonstrate clearly how the system violates the property. For each system, define MATLAB vectors representing the input(s) and output(s). Then, make plots of these signals, and construct a well reasoned argument explaining how these figures demonstrate that the system fails to satisfy the property in question.

(a). The system $y[n] = \sin((\pi/2)x[n])$ is not linear. Use the signals $x_1[n] = \delta[n]$ and $x_2[n] = 2\delta[n]$ to demonstrate how the system violates linearity.

(b). The system $y[n] = x[n] + x[n+1]$ is not causal. Use the signal $x[n] = u[n]$ to demonstrate this. Define the MATLAB vectors **x** and **y** to represent the input on the interval $-5 \leq n \leq 9$, and the output on the interval $-6 \leq n \leq 9$, respectively.

Code Logic:

xn = (nn >= 0);: Using the definition of the unit step signal u[n], it directly obtains the current input value x[n] by evaluating the condition nn >= 0 (returns 1 if true, 0 if false).

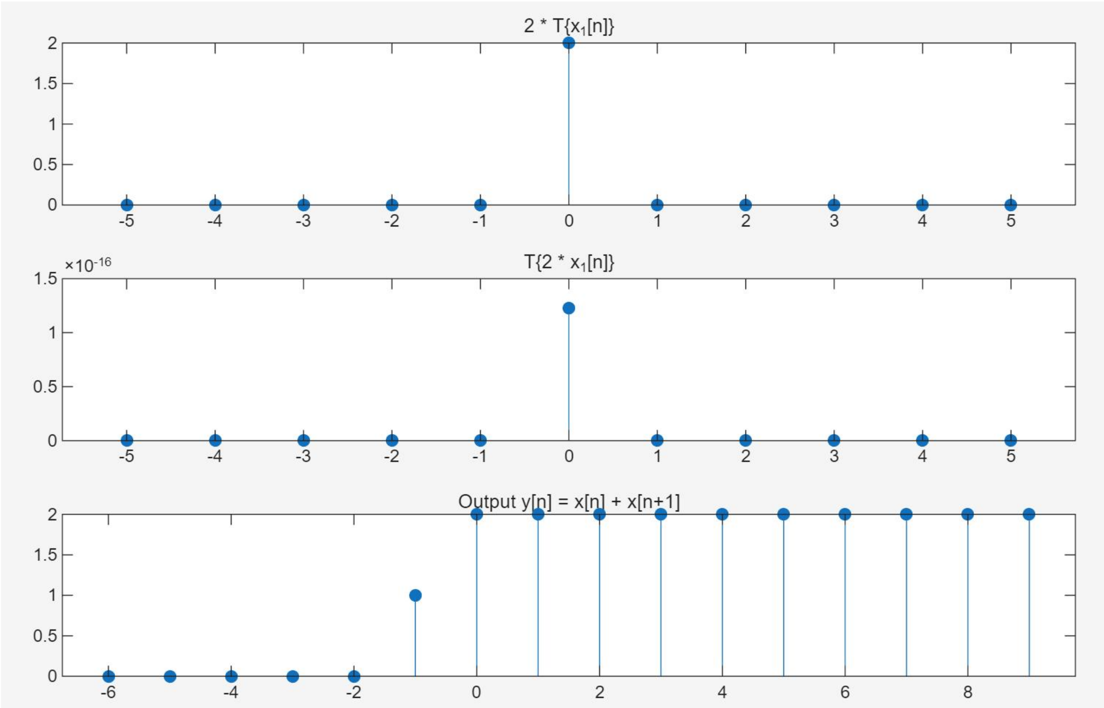
xn_plus_1 = (nn+1 >= 0);: Similarly, it directly acquires the look-ahead (future) input value x[n+1] required by the system by evaluating nn+1 >= 0.

```
clear; clc;
% 非线性证明
n = -5:5;
delta = (n == 0);            % x1[n] = delta[n]
x2 = 2 * delta;

y1 = sin((pi/2) * delta);
y2 = sin((pi/2) * x2);

figure;
subplot(3, 1, 1); stem(n, 2*y1, 'filled'); title('2 * T\{x_1[n]\}');
subplot(3, 1, 2); stem(n, y2, 'filled'); title('T\{2 * x_1[n]\}');

% 非因果证明
n_in = -5:9;
x = (n_in >= 0);             %  x[n] = u[n]
n_out = -6:9;
y = zeros(size(n_out));

for idx = 1:length(n_out)
    nn = n_out(idx);
    xn = (nn >= 0);
    xn_plus_1 = (nn+1 >= 0);
    y(idx) = xn + xn_plus_1;
end

subplot(3, 1, 3);
stem(n_out, y, 'filled');
title('Output y[n] = x[n] + x[n+1]'); xlabel('n');
```

Experimental Results and Conclusion:

System (a) is non-linear: Based on the code calculations and plotted graphs, when the input is x1[n]=delta[n], the output is y1[n]=delta[n]. If the system were linear, scaling the input by a factor of 2 should proportionally scale the output by 2 . However, $T\{2x1[n]\} \neq 2T\{x1[n]\}$, the system fails to satisfy the principle of homogeneity, robustly proving it is a non-linear system.

System (b) is non-causal: Observing the graph plotted by MATLAB, when the input is a unit step sequence u[n], the system's output value at n=-1$ is y[-1] = 1. The definition of a causal system requires that the output at a current time can only depend on current or past inputs. Because this system's output at the current time (n=-1) depends in advance on the input state at a future time, proving the system is non-causal.



Problem 6

(a). Write a function **y=diffeqn(a,x,yn1)** which computes the output $y[n]$ of the causal system determined by Eq. (1.6). The input vector **x** contains $x[n]$ for $0 \le n \le N-1$ and **yn1** supplies the value of $y[-1]$. The output vector **y** contains $y[n]$ for $0 \le n \le N-1$. The first line of your M-file should read

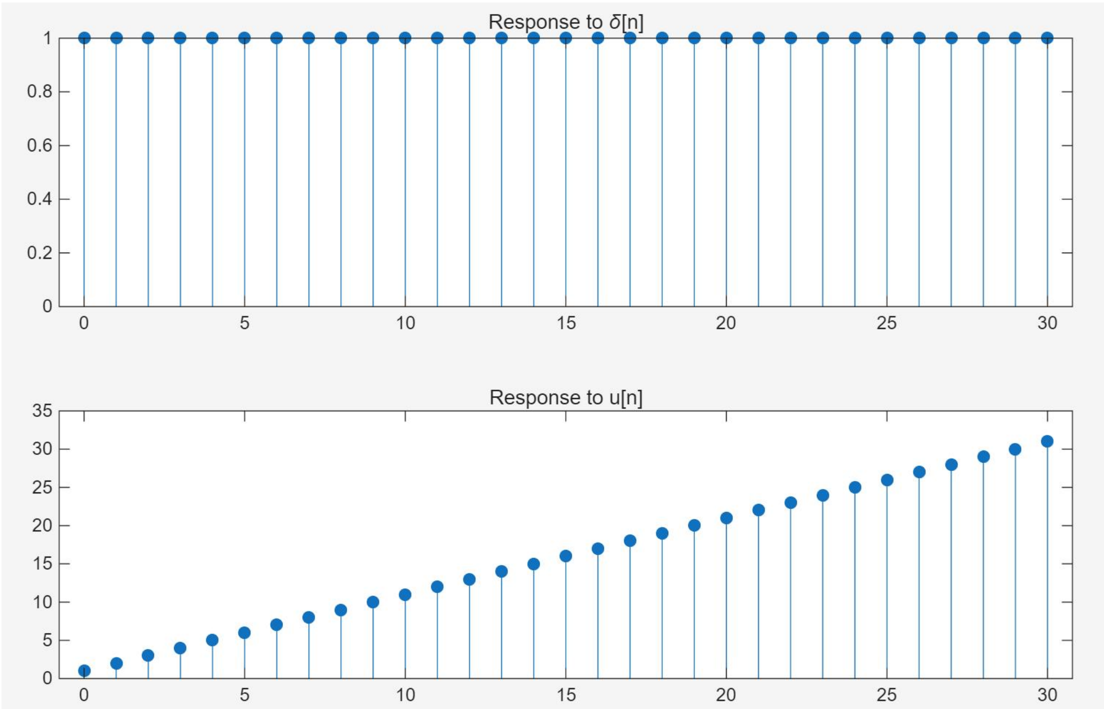
```
function y = diffeqn(a,x,yn1)
```

Hint: Note that $y[-1]$ is necessary for computing $y[0]$, which is the first step of the autoregression. Use a **for** loop in your M-file to compute $y[n]$ for successively larger values of n, starting with $n = 0$.

(b). Assume that $a = 1$, $y[-1] = 0$, and that we are only interested in the output over the interval $0 \le n \le 30$. Use your function to compute the response due to $x_1[n] = \delta[n]$ and $x_2[n] = u[n]$, the unit impulse and unit step, respectively. Plot each response using **stem**.

```
% diffeqn.m
function y = diffeqn(a, x, yn1)
    N = length(x);
    y = zeros(1, N);
    y(1) = a * yn1 + x(1);

    for n = 2:N
        y(n) = a * y(n-1) + x(n);
    end
end
```

```
clear; clc;
% Problem 6 (b)
n = 0:30;

x1 = (n == 0);  % delta[n]
x2 = (n >= 0);  % u[n]

y1 = diffeqn(1, x1, 0);
y2 = diffeqn(1, x2, 0);

figure;
subplot(2, 1, 1); stem(n, y1, 'filled'); title('Response to \delta[n]');
subplot(2, 1, 2); stem(n, y2, 'filled'); title('Response to u[n]');
```



Experimental Results and Conclusion:

Impulse Response Verification: When the unit impulse signal x1[n] = delta[n] is input into the system, the output becomes a sequence of all 1, which is the unit step sequence u[n].

Step Response Verification: When the unit step signal x2[n] = u[n] is input, the system continuously accumulates the '1's input at each time step. The output exhibits a step-like linear growth, forming a ramp sequence.

Under this specific parameter, the difference equation is equivalent to an "integrator" in discrete-time systems, which performs discrete-time accumulation on the input signal.

As shown by the delta[n] test results, even if the external stimulus only exists for a brief instant at n=0, a system with feedback can generate a persistent and infinitely long response output.

Problem 7

Consider the continuous-time sinusoid

$$x(t) = \sin(2\pi t/T)\,.$$

A symbolic expression can be created to represent $x(t)$ within MATLAB by executing

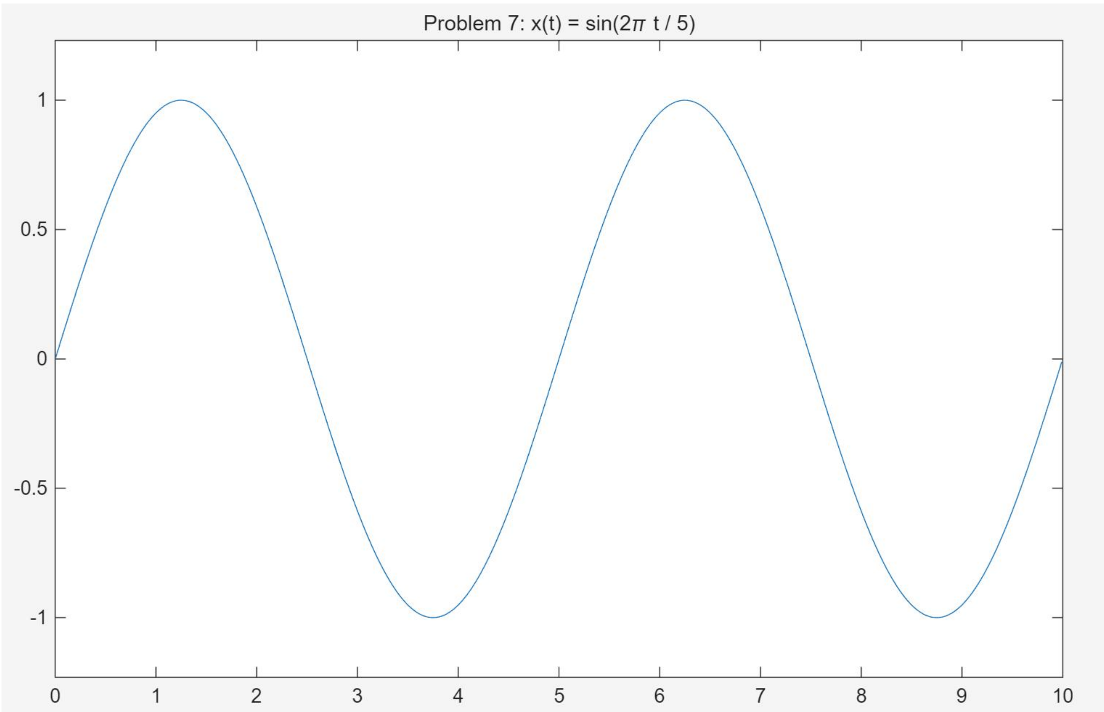
```
>>  x = sym('sin(2*pi*t/T)');
```

The variables of **x** are the single character strings **'t'** and **'T'**. The function **ezplot** can be used to plot a symbolic expression which has only one variable, so you must set the fundamental period of $x(t)$ to a particular value. If you desire $T = 5$, you can use **subs** as follows

```
>>  x5 = subs(x,5,'T');
```

Thus **x5** is a symbolic expression for $\sin(2\pi t/5)$. Create the symbolic expression for **x5** and use **ezplot** to plot two periods of $\sin(2\pi t/5)$, beginning at $t = 0$. If done correctly, your plot should be as shown in Figure 1.2.

```
clear; clc;
syms t T

x = sin(2*pi*t/T);
x5 = subs(x, T, 5);

figure;
ezplot(x5, [0, 10]);
title('Problem 7: x(t) = sin(2\pi t / 5)');
```



Experimental Results and Conclusion:

Different from the stem plots (stem) used for discrete-time sequences in previous problems, the symbolic plotting function (ezplot) generates an extremely smooth continuous sine curve with no breakpoints. This visually perfectly simulates the true morphology of continuous-time (CT) signals in the physical world.

This experiment demonstrates that with the help of MATLAB's Symbolic Math Toolbox, we can temporarily bypass the inherent limitations of "sampling rate" and "discrete step size" of digital computers when analyzing continuous systems.

Problem 8

In the following problems you will write M-files for extracting the real and imaginary components, or the magnitude and phase, of a symbolic expression for a complex signal.

(d). Store in **x** a symbolic expression for the signal

$$x(t) = e^{i2\pi t/16} + e^{i2\pi t/8}.$$

Remember to use **'i'** rather than **'j'** within symbolic expressions to represent $\sqrt{-1}$. The function **ezplot** cannot be used directly for plotting $x(t)$, since $x(t)$ is a complex signal. Instead, the real and imaginary components must be extracted and then plotted separately.

(e). Write a function **xr=sreal(x)** which returns a symbolic expression **xr** representing the real part of $x(t)$. If your function is working properly, **ezplot(xr)** will plot the real component of $x(t)$. Similarly, write a function **xi=simag(x)** which returns a symbolic expression **xi** representing the imaginary component of $x(t)$. The first line of the M-file **sreal.m** should be

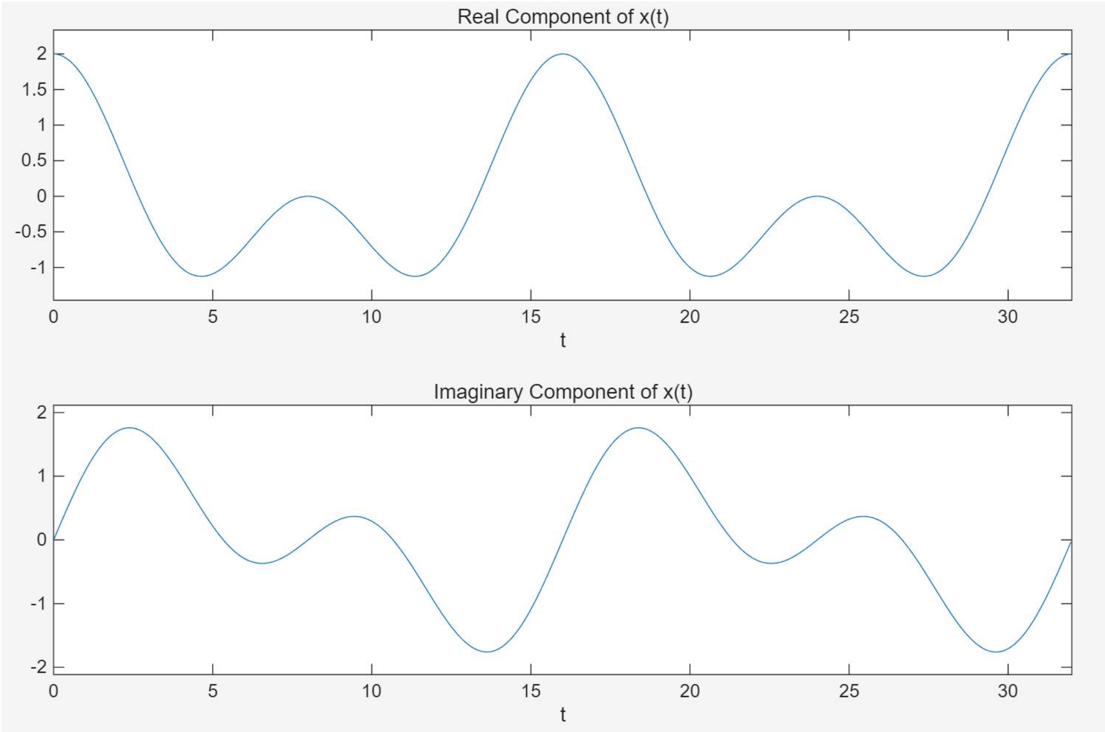
function xr = sreal(x)

You can then use **compose('real(x)',x)** to create a symbolic expression for the real component of $x(t)$. Use **ezplot** and the functions you created to plot the real and imaginary components of $x(t)$ on the interval $0 \leq t \leq 32$. Use a separate plot for each component. What is the fundamental period of $x(t)$?

```
% sreal.m
function xr = sreal(x)
    xr = real(x);
end
```

```
% simag.m
function xi = simag(x)
    xi = imag(x);
end
```

```
clear; clc;
syms t

x_complex = exp(1i*2*pi*t/16) + exp(1i*2*pi*t/8);
xr = sreal(x_complex);
xi = simag(x_complex);

figure;
subplot(2, 1, 1);
ezplot(xr, [0, 32]);
title('Real Component of x(t)');

subplot(2, 1, 2);
ezplot(xi, [0, 32]);
title('Imaginary Component of x(t)');
```



Experimental Results and Conclusion:

$$\mathrm{Real}\{x(t)\} = \cos(\tfrac{2\pi t}{16}) + \cos(\tfrac{2\pi t}{8})$$
$$\mathrm{Imag}\{x(t)\} = \sin(\tfrac{2\pi t}{16}) + \sin(\tfrac{2\pi t}{8})$$

The two generated images perfectly display these cosine and sine waveform characteristics.

The first component $e^{i\frac{2\pi t}{16}}$: T1 = 16.

The second component $e^{i\frac{2\pi t}{8}}$: T2 = 8

The fundamental period of the sum of two periodic signals is the least common multiple of their respective periods. The least common multiple of 16 and 8 is 16. Therefore, the overall fundamental period of this complex signal is T = 16.